

Sheet 5

1. $y_t = \beta t + \varepsilon_t$, $\varepsilon_t \stackrel{iid}{\sim} (0, \sigma_\varepsilon^2)$, $\mu_\varepsilon < \infty$

a) $E[y_t] = \beta t + E[\varepsilon_t]$ by linearity
 $= \beta t$

1a. correct!

$\text{Var}(y_t) = \text{var}(\varepsilon_t) = \sigma_\varepsilon^2$ since βt is non-random

$\text{Cov}(y_t, y_{t-h}) := E[(y_t - E[y_t])(y_{t-h} - E[y_{t-h}])]$

$= E[(\varepsilon_t)(\beta_{t-h} + \varepsilon_{t-h} - \beta_{t-h})]$ by $E[\varepsilon_t] = 0$
 $= E[\varepsilon_t \varepsilon_{t-h}] = 0$ since $\varepsilon_i \perp \varepsilon_j \forall i \neq j$ and mean zero

$\text{corr}(y_t, y_{t-h}) := \gamma_h / \sqrt{\text{var}(y_t) \text{var}(y_{t-h})} = 0$

b) No, definitely not, because the expectation $E[y_t]$ is a function of time βt , so the mean fails to be time-invariant, unless $\beta = 0$.
 However, note that the first difference $\Delta y_t := y_t - y_{t-1} = \beta - \varepsilon_{t-1} + \varepsilon_t$ would be weakly stationary (mean β , variance $2\sigma_\varepsilon^2$, zero autocorrelation). 1b. yes, correct!

c) Consider the first difference Δy_t defined above, and let $m_t := \Delta y_t - \beta$. Then $E[m_t | I_{t-1}] = E[-\varepsilon_{t-1} + \varepsilon_t | I_{t-1}]$. [Actually, not an mds?]

Consider the ~~sample mean~~ ^{expression} $\frac{1}{T^2} \sum_{t=1}^T (\beta t + \varepsilon_t) = \underbrace{\beta \frac{1}{T^2} \sum_{t=1}^T t}_{(1)} + \underbrace{\frac{1}{T} \frac{1}{T} \sum_{t=1}^T \varepsilon_t}_{(2)}$

(1) $\rightarrow \beta \int_0^1 \frac{dr}{dr} = \frac{1}{2} \beta$

1c. ok, correct

(2) By the Chebyshev LLN, since ε_t are iid with $\sigma_\varepsilon^2 < \infty$, $\frac{1}{T} \sum \varepsilon_t \xrightarrow{P} E[\varepsilon_t] = 0$. And $\lim_{T \rightarrow \infty} \frac{1}{T} = 0$, so (2) $\xrightarrow{P} 0$. (Or becomes degenerate?)

So normalising by T^2 we have a kind of LLN. Clearly if we just take the sample mean of y_t there is no LLN applicable since $E[y_t] \rightarrow \pm \infty$, given the deterministic trend $\neq 0$.

d) $\hat{\beta} = \arg \min_{b \in \mathbb{R}} \sum_{t=1}^T (y_t - bt)^2$ for which the FOC is $\frac{\partial L}{\partial b} = 0$

1d. yes, correct

$\therefore -2 \sum_{t=1}^T t(y_t - \hat{\beta}t) = 0$

$\Rightarrow \sum_{t=1}^T t y_t - \hat{\beta} \sum_{t=1}^T t^2 = 0$

$\hat{\beta} = \frac{\sum_{t=1}^T t y_t}{\sum_{t=1}^T t^2} = \frac{\sum_{t=1}^T (\beta t^2 + \varepsilon_t t)}{\sum_{t=1}^T t^2} = \beta + \frac{\sum_{t=1}^T \varepsilon_t t}{\sum_{t=1}^T t^2}$

This is just like a slope-only regression where our regressor is t .

e) Consistency requires that $\hat{\beta} \xrightarrow{P} \beta \Leftrightarrow (\hat{\beta} - \beta) \xrightarrow{P} 0$.

$\hat{\beta} - \beta = \frac{\sum \varepsilon_t t}{\sum t^2}$ Consider first the denominator, which grows at rate T^3

Note that the denominator $\sum t^2$ grows at rate T^3 .

A sufficient condition for this is convergence in mean square.

$E[\hat{\beta}] = \beta + E\left[\frac{\sum \varepsilon_t t}{\sum t^2}\right] = \frac{\sum t E[\varepsilon_t]}{\sum t^2}$

$= \beta + 0$ since $E[\varepsilon_t] = 0$ and $\sum t^2 < \infty$.

so $\hat{\beta}$ is unbiased, $E[\hat{\beta} - \beta] = 0$

1e. yes, correct!

Thus $E[(\hat{\beta} - \beta)^2] = \text{Var}(\hat{\beta} - \beta)$. So if we can show $\lim_{T \rightarrow \infty} \text{Var}(\hat{\beta} - \beta) = 0$, we'll have consistency.

$\text{var}(\hat{\beta} - \beta) = \frac{1}{(\sum t^2)^2} \text{var}(\sum \varepsilon_t t) = (\sum t^2)^{-2} \sum t^2 \text{var} \varepsilon_t$ since ε_t iid σ^2

Let $S_T = \sum t^2$. Then $\text{var}(\hat{\beta} - \beta) = (S_T)^{-2} \cdot \sigma_\varepsilon^2 \cdot S_T$

$= \sigma_\varepsilon^2 / S_T$. And the denominator S_T grows at rate T^3 ,

while the numerator is constant as $T \rightarrow \infty$. So

indeed $\text{var}(\cdot) \rightarrow 0$, and $\Rightarrow$ convergence in mean square

$\Rightarrow$ consistency. (In fact, we have super consistency

since $\text{var}(\cdot)_n$ decreases at rate T^3 , not just T^1 .)

f) Recall that $\hat{\beta} = \beta + \frac{\sum \epsilon_t t}{\sum t^2}$.

Let us consider the standardised estimator $T^{3/4}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{T}} \sum \epsilon_t t / T}{\frac{1}{T} \sum (t/T)^2}$.

We know that the denominator $\rightarrow \int_0^1 r^2 dr = \frac{1}{3}$, as in (c).

Turning now to the numerator, let $m_t := \frac{t}{T} \epsilon_t$. We want to apply the mds CLT, which requires that

(i) $\sum m_t^2 / S_T^2 \xrightarrow{P} 1$ and

(ii) $\sum E[|m_t| S_T^{2+\delta}] \rightarrow 0$ for some $\delta > 0$,

where $S_T^2 := \sum E[m_t^2]$ and m_t is an mds.

Since $\epsilon_i \perp \epsilon_j, \forall i \neq j$, we have $E[\epsilon_t | \{\epsilon_{t-1}\}] = E[\epsilon_t]$, and thus $E[m_t | \mathcal{I}_{t-1}] = \frac{t}{T} E[\epsilon_t] = 0$, so m_t is an mds, as required.

In this case, $S_T^2 = \sum \text{var}(m_t) = \sigma_\epsilon^2 \sum (t/T)^2$ since ϵ_t ident. dist.
 $= \sigma_\epsilon^2 / T^2 \sum t^2$

(ii). Let $\delta = 2$. RTP:

$$\sum \frac{(t/T)^4 E[\epsilon_t^4]}{S_T^4} \rightarrow 0.$$

$$\Leftrightarrow \frac{\mu_4}{S_T^4 / T} \cdot \frac{1}{T} \sum \left(\frac{t}{T}\right)^4 \rightarrow 0 \quad \text{since } E[\epsilon_t^4] = \mu_4 < \infty$$

$$\Rightarrow \frac{\mu_4}{T^3 \sum t^2} \rightarrow \frac{1}{5} \text{ by previous result.}$$

$\rightarrow \infty$ as $(\sum t^2)^2$ is $O(T^6)$ So as required the overall fraction $\rightarrow 0$.

(iii)

~~(i) We can use the mds LLN, which requires it's sufficient to show $\lim_{T \rightarrow \infty} \max_{t \leq T} E[q_t^p]^{1/p} < \infty$ for some $p \in [0, 2]$.~~

Not our condition (i) $\Leftrightarrow \frac{1}{T} \sum m_t^2 - S_T^2 / T \xrightarrow{P} 0$.

$$\Leftrightarrow \frac{1}{T} \cdot \sum \left(\frac{t}{T}\right)^2 (\epsilon_t^2 - E[\epsilon_t^2]) \xrightarrow{P} 0 \quad \text{by def'n of } S_T^2.$$

Observe that $q_t := (\frac{t}{T})^2 (\epsilon_t^2 - E[\epsilon_t^2])$ is an mds, because

RSR: probably a slip with powers here but argument correct overall.

$$E[q_t | I_{t-1}] = 0, \text{ from } \epsilon_t \text{ all independent.}$$

We want to apply an mds LLN. A sufficient condition to do so on q_t is that $\lim_{T \rightarrow \infty} \max_{t \leq T} E[q_t^{1+p}] < \infty$ for some $p \in [0, 2]$.

Letting $p=2$, then the inner portion $E[q_t^2] = \underbrace{(t/T)^4}_{\in [0,1]} \underbrace{E[\epsilon_t^2 - E[\epsilon_t^2]]^2}_{\text{some constant } C \ll t, \text{ since } \mu_4 < \infty}$

1f. excellent answer!

So even as $T \rightarrow \infty$ that max will be $< \infty$.

Applying the LLN, we have $T^{-1} \sum q_t \xrightarrow{P} 0$ exactly as needed.

So, (i) and (ii) are satisfied. We can use the mds CLT:

$$\frac{1}{S_T} \sum m_t \xrightarrow{d} N(0, 1). \Leftrightarrow \frac{1}{S_T \sqrt{T}} \sum m_t \xrightarrow{d} N(0, 1)$$

$\therefore$ Our numerator $\xrightarrow{d} N(0, S_T^2/T)$ and the denominator $\xrightarrow{P} \frac{1}{3}$.

$$\text{Thus } T^{3/2} (\hat{\beta} - \beta) \xrightarrow{d} N(0, 3S_T^2/T)$$

But since $S_T^2/T = \frac{1}{T} \sum (t/T)^2 E[\epsilon_t^2] = \sigma_\epsilon^2 \frac{\sum t^2}{T^3} \rightarrow \sigma_\epsilon^2 \cdot \frac{T^3}{T^3}$,
we can write

$$T^{3/2} (\hat{\beta} - \beta) \xrightarrow{d} N(0, 3\sigma_\epsilon^2)$$

g) a. $E[y_t] = \beta t^2$, $\text{Var}(y_t) = \sigma_\epsilon^2$, $\gamma_h = \rho_h = 0$

b. No, not weakly stationary.

c. Again, there's a kind of LLN; we just now have to normalize by T^3 and we can have (1) $= \beta \frac{1}{T} \sum (t/T)^2 \rightarrow \frac{1}{3} \beta$, (2) $= \frac{1}{T^2} \sum \epsilon_t \xrightarrow{P} 0$. But as before $E[y_t] \rightarrow \pm \infty$ given $\beta \neq 0$ so there's not a straightforward LLN.

d. $\hat{\beta} = \arg \min_{b \in \mathbb{R}} \sum (y_t - bt^2)^2$, FOC $\frac{\partial L}{\partial b} = 0$

$$\therefore \hat{\beta} = \frac{\sum t^2 y_t}{\sum t^4} = \beta + \frac{\sum \varepsilon_t t^2}{\sum t^4}$$

i.e. we just let the "regressor" be t^2 rather than t .

e. Yes. First, we know as before that $\hat{\beta}$ is unbiased.

And second, we again have mean square convergence: ~~mean~~
 $\lim_{T \rightarrow \infty} \text{var}(\hat{\beta} - \beta) = 0$, using exactly the argument as previously except replacing t with t^2 . So the variance falls to zero even faster than before, at T^{-5} , since $S_T' := \sum t^4$.

1g. yes, all correct, although you should check all the conditions for the MDS CLT hold in this case as well

+ using the same arguments, we can show that

$$T^{5/2}(\hat{\beta} - \beta) \overset{d}{\rightarrow} N(0, 5\sigma_\varepsilon^2).$$

2. $y_t = \alpha + \delta t + \varepsilon_t$, $\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

2a. yes, correct

a) $E[y_t] = \alpha + \delta t + E[\varepsilon_t] = \alpha + \delta t$ by linearity
 $\text{var}(y_t) = \text{var}(\varepsilon_t) = \sigma_\varepsilon^2$ since α, δ, t non-random

$\text{cov}(y_t, y_{t-h}) = \text{cov}(\alpha + \delta t + \varepsilon_t, \alpha + (t-h)\delta + \varepsilon_{t-h})$
 $= \text{cov}(\varepsilon_t, \varepsilon_{t-h}) = 0$ since ε_t independent and rest non-random

$\rho_h = \gamma_h / \sqrt{\text{var}(y_t) \text{var}(y_{t-h})} = 0$

b) No, because the mean isn't time-invariant; instead $E[y_t]$ is a linear function of t . 2b. yes, correct

c) As in 1c), if we scale by T^{-2} , then $T^{-2} \sum y_t = T^{-2} \sum \alpha + \delta t + \varepsilon_t$

2c. ok, but without the

proposed scaling it is...?

$$= \underbrace{T^{-1} \alpha}_{\rightarrow 0} + \underbrace{T^{-1} \delta \sum \frac{t}{T}}_{\rightarrow \frac{1}{2} \delta \text{ by hint}} + \underbrace{T^{-1} \cdot T^{-1} \sum \varepsilon_t}_{\rightarrow 0 \text{ by Chebyshev LLN}}$$

d) Let $\hat{\beta} := (\hat{\alpha}, \hat{\delta})' = \underset{\beta \in \mathbb{R}^2}{\operatorname{argmin}} \sum (y_t - x_t \beta)' (y_t - x_t \beta)$

with FOC $\frac{\partial L}{\partial \beta} \stackrel{!}{=} 0$ where $L := Y'Y - 2\beta'X'Y + \beta'X'X\beta$

and $Y := (y_1, \dots, y_T)'$

$X := \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 2 & \dots & T \end{pmatrix}'$

$\therefore -2X'Y + 2X'X\hat{\beta} = 0 \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$

Since $\operatorname{rank}(X) = 2$, so too is $\operatorname{rank}(X'X)$ and thus it's invertible (full-rank)

$= (X'X)^{-1}X'(X\beta + U)$

Since $Y = X\beta + U$ where $U := (\varepsilon_1, \dots, \varepsilon_T)'$

$= \beta + (X'X)^{-1}X'U$ by distributive property, where $\beta := (\alpha, \delta)'$

$X'X$ is a the 2×2 symmetric matrix $\begin{pmatrix} T & \sum t \\ \sum t & \sum t^2 \end{pmatrix}$

2d. yes, correct

so $\hat{\beta} = \begin{pmatrix} T & \sum t \\ \sum t & \sum t^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum y_t \\ \sum t y_t \end{pmatrix} = \beta + \begin{pmatrix} T & \sum t \\ \sum t & \sum t^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum \varepsilon_t \\ \sum t \varepsilon_t \end{pmatrix}$

e) ~~But~~ $\hat{\beta} \xrightarrow{P} \beta$ is ~~iterative~~ equivalent to $\hat{\beta} - \beta \xrightarrow{P} 0$, so it will suffice to show that $(X'X)^{-1}X'U \xrightarrow{P} 0$.

~~XXXXXX~~ ~~(X'X)^{-1}~~ Note that the entries in $(X'X)^{-1}$ grow at rates $\begin{pmatrix} T^{-1} & T^{-2} \\ T^{-2} & T^{-3} \end{pmatrix}$.

RSR: better to write $\sim$, not $\rightarrow$

so $\hat{\beta} - \beta \rightarrow \begin{pmatrix} \frac{1}{T} \sum \varepsilon_t + \frac{1}{T} \sum t \varepsilon_t \\ \frac{1}{T} \sum t \varepsilon_t + \frac{1}{T} \cdot \frac{1}{T} \sum t^2 \varepsilon_t \end{pmatrix}$ how...?

As in (c), applying limits gives us $\frac{1}{T} \sum \varepsilon_t \xrightarrow{P} 0$. Since $t/T \in [0, 1]$, $\frac{1}{T} \sum t \varepsilon_t$ is ~~upper~~ bounded by that and also $\xrightarrow{P} 0$.

And the final term ~~for the~~ is equal to $\frac{1}{T} \sum (t/T) \cdot \frac{1}{T} \sum \varepsilon_t \xrightarrow{P} \frac{1}{2} \cdot 0 = 0$

So as required $\hat{\beta} - \beta \rightarrow 0$ and we have consistency.

2e. ok, correct final conclusion but not sure i follow the argument...(could be correct, so please verify with the tutor)

You could show element-wise convergence in mean square, but what is the best matrix approach? It seemed like scaling with a diagonal matrix doesn't necessarily give us the required result.

But I don't think my argument works very well either? (In cross-sectional context we took $(X'X/n)^{-1}(X'U/n)$ and analysed that, but scalar scaling doesn't work here I think.

e) $\hat{\beta} \xrightarrow{P} \beta \Leftrightarrow \hat{\beta} - \beta \xrightarrow{P} 0$, so we can demonstrate consistency by showing $(X'X)^{-1}X'U \xrightarrow{P} 0$.

Consider first $(X'X)^{-1} = \begin{pmatrix} T & \sum t \\ \sum t & \sum t^2 \end{pmatrix}^{-1}$. In the limit, we know that this $\rightarrow \begin{pmatrix} T & T^2 \\ T^2 & T^3 \end{pmatrix}^{-1} \approx \begin{pmatrix} T^{-1} & T^{-2} \\ T^{-2} & T^{-3} \end{pmatrix}$.

Now examine $X'U = \begin{pmatrix} \sum \varepsilon_t \\ \sum t \varepsilon_t \end{pmatrix}$. From before

f) Let $D = \begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix}$ be our scaling matrix, and now consider the distribution of the standardised estimator

$$D(\hat{\beta} - \beta) = \underbrace{(D^{-1}X'XD^{-1})^{-1}}_{M_4} \underbrace{D^{-1}X'U}_{M_5P}$$

Start with M_4 's limiting form. $D^{-1} = \frac{1}{T^2} \begin{pmatrix} T^{3/2} & 0 \\ 0 & T^{1/2} \end{pmatrix} = \begin{pmatrix} T^{-1/2} & 0 \\ 0 & T^{-3/2} \end{pmatrix}$

$$\therefore M_4^{-1} = \begin{pmatrix} T^{-1} \cdot T & T^{-2} \cdot \sum t \\ T^{-2} \cdot \sum t & T^{-3} \cdot \sum t^2 \end{pmatrix}^{-1} \rightarrow \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}^{-1} \text{ by Riemann sums.}$$

$\therefore Q^{-1}$

Now examine $M_5P = \begin{pmatrix} T^{-1/2} \sum \varepsilon_t \\ T^{-1/2} \sum (t/T) \varepsilon_t \end{pmatrix}$. By the ~~central limit theorem~~, since ε_t is iid with finite second moment, $\frac{1}{\sqrt{T}} \sum \varepsilon_t \rightarrow N(0, \sigma^2)$.

By the Lindeberg-Lévy CLT, since ε_t is iid with at least finite second moment, $P_1 = \frac{1}{\sqrt{T}} \sum \varepsilon_t \xrightarrow{d} N(0, \sigma^2)$.

As for $P_2 = \frac{1}{\sqrt{T}} \sum (t/T) \varepsilon_t$, note first that $E[\varepsilon_t | I_{t-1}] = 0$.

$$E[(t/T) \varepsilon_t | I_{t-1}] = t/T \cdot E[\varepsilon_t | I_{t-1}] = 0, \text{ so}$$

by independence of $\{\varepsilon_t\}_t$

$u_t := (t/T) \cdot \varepsilon_t$ is an mds.

As in Q1f), we can apply the mds CLT to u_t , showing that the conditions are satisfied by appeal to the mds LLN. The result that $P_2 \xrightarrow{d} N(0, \sigma^2 I_3)$ will follow as before.

But we now need to consider the joint distribution of P .

Letting $\lambda = (\lambda_1, \lambda_2)' \in \mathbb{R}^2$, we can apply the Cramer-Wold device: since $\lambda' P \xrightarrow{d} \lambda' N(0, \sigma^2 Q)$ by the mds CLT on the weighted sum for all $\lambda \in \mathbb{R}^2$,

2f. yes, correct!

we have that $P \xrightarrow{d} N(0, \sigma^2 Q)$.

recall
And $D(\hat{\beta} - \beta) = M'P$, with $M' \rightarrow Q^{-1}$, $P \xrightarrow{d} N(0, \sigma^2 Q)$.

so $D(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1})$ by variance rules.

g) To test $H_0: \alpha = 0$, we calculate the t -statistic
 $t_\alpha = \hat{\alpha} / \sqrt{\hat{\text{var}}(\hat{\alpha})}$, where $\hat{\text{var}}(\hat{\alpha}) = s^2 (X'X)^{-1}_{11}$
and s^2 is the ~~sample~~ error variance estimator
 $\frac{1}{T-2} (Y - X\hat{\beta})'(Y - X\hat{\beta})$

2g. yes, correct!

similarly $t_\delta = \hat{\delta} / \sqrt{\hat{\text{var}}(\hat{\delta})}$ where $\hat{\text{var}}(\hat{\delta}) = s^2 (X'X)^{-1}_{22}$.

By construction and given 2f), in large samples the t -statistic will $\approx N(0, 1)$ under H_0 , so we can perform inference as usual, rejecting H_0 if $|t^{\text{obs}}| > c_\alpha$ for some significance level α .

Since our errors are normal iid, I think we can even do exact inference in finite samples, with $t \sim t_{T-2}$.
(This is true in the cross-sectional setting but I'm unsure if it **yes, should be right but check with tutor :)** series.)